

Regna — Monorepo setup (pnpm, Turborepo, Nuxt 4, Fastify)

Corrected guide aligned with this repository. French version of the visual doc: `docs/HTML text.html` .

1. Prerequisites — install pnpm

```
npm install -g pnpm
pnpm --version
```

Corepack is also valid: `corepack enable && corepack prepare pnpm@11.1.3 --activate` .

2. Root monorepo

```
mkdir regna && cd regna
pnpm init
pnpm add -D turbo typescript
mkdir -p packages/engine apps/client apps/server
```

pnpm-workspace.yaml

```
packages:
  - "packages/*"
  - "apps/*"

# Optional: allow native installs for some dependencies (matches this repo)
allowBuilds:
  "@parcel/watcher": true
  esbuild: true
```

turbo.json (Turborepo 2.x uses tasks , not pipeline)

```
{
  "tasks": {
    "build": {
      "dependsOn": ["^build"],
      "outputs": ["dist/**"]
    },
    "dev": {
      "cache": false,
      "persistent": true
    },
    "typecheck": {}
  }
}
```

```
}  
}
```

Root `package.json` essentials

```
{  
  "name": "Regna",  
  "packageManager": "pnpm@11.1.3",  
  "scripts": {  
    "dev": "turbo run dev",  
    "build": "turbo run build",  
    "typecheck": "turbo run typecheck"  
  },  
  "devDependencies": {  
    "turbo": "^2.9.14",  
    "typescript": "^6.0.3"  
  }  
}
```

Turborepo + pnpm: Do **not** combine the root `packageManager` field with `devEngines.packageManager` in the same `package.json`. Turborepo reports that it will ignore `packageManager` and workspace resolution can fail (e.g. "Missing `packageManager` field" / "Could not resolve workspaces"). Use **only** `"packageManager": "pnpm@..."`.

3. `packages/engine` — shared rules engine

```
cd packages/engine  
pnpm init  
pnpm add zod  
pnpm add -D typescript  
npx tsc --init --strict --target ES2022 --module NodeNext --moduleResolution  
NodeNext --outDir dist
```

`packages/engine/package.json` (important)

The **name** field must match workspace dependencies (e.g. `"@chess/engine"`), not only the folder name `engine`.

```
{  
  "name": "@chess/engine",  
  "version": "1.0.0",  
  "type": "module",  
  "main": "./dist/index.js",  
  "scripts": {  
    "dev": "tsc --watch",  
    "build": "tsc"  
  },  
  "dependencies": {
```

```
    "zod": "^4.4.3"
  },
  "devDependencies": {
    "typescript": "^6.0.3"
  }
}
```

Pure TypeScript, no UI or Fastify here; consumed by both `apps/client` and `apps/server` .

4. `apps/server` — Fastify + Socket.IO

```
cd apps/server
pnpm init
pnpm add fastify fastify-plugin @fastify/cors socket.io drizzle-orm postgres zod
pnpm add "@chess/engine@workspace:*"
pnpm add -D typescript tsx @types/node drizzle-kit
```

Example excerpt:

```
{
  "name": "server",
  "type": "module",
  "scripts": {
    "dev": "tsx watch src/index.ts",
    "build": "tsc"
  },
  "dependencies": {
    "@chess/engine": "workspace:*"
  }
}
```

5. `apps/client` — Nuxt 4 + PixiJS

This repo uses **Nuxt 4** (not Nuxt 3).

```
pnpm dlx nuxi@latest init apps/client
# In prompts: minimal, pnpm; avoid duplicating a nested git repo if you already have
one at monorepo root.
```

If pnpm blocks postinstall builds:

```
pnpm approve-builds
```

Dependencies (align with current app):

```
cd apps/client
pnpm add pixi.js @vueuse/core vue vue-router "@chess/engine@workspace:*
```

```
pnpm add -D @vite-pwa/nuxt
pnpm add nuxt@^4
```

nuxt.config.ts (minimal, as in repo)

```
export default defineNuxtConfig({
  modules: ['@vite-pwa/nuxt'],
  pwa: {
    manifest: {
      name: 'Regna',
      short_name: 'Regna',
      theme_color: '#3D5AFE'
    }
  }
})
```

PWA: Do not use `@nuxtjs/pwa` — it is unmaintained and not appropriate for Nuxt 4; use `@vite-pwa/nuxt` .

App package naming in this repo: `"name": "client"` with `"private": true` is fine.

6. Final install & dev

From the repository root (where `pnpm-workspace.yaml` lives):

```
pnpm install
pnpm dev
```

Run only the client:

```
cd apps/client && pnpm dev
```

pnpm: Prefer `pnpm install` from the root so every workspace package links consistently. Installing from a subpackage still resolves the workspace root in many setups; the important part is keeping all `package.json` files coherent.

7. Expected layout

```
Regna/
├─ packages/
│   └─ engine/           # npm name: @chess/engine
├─ apps/
│   └─ server/
│   └─ client/           # Nuxt 4 + PixiJS
├─ pnpm-workspace.yaml
├─ turbo.json
└─ package.json
```

Changelog vs old HTML draft

Issue	Correction
@regna/engine vs real package	Workspace package is @chess/engine .
@regna/client / @regna/server	Apps use client and server .
Nuxt "3"	Stack is Nuxt 4 .
turbo.json	Use tasks , declare typecheck if the root runs <code>turbo run typecheck</code> .
Turborepo error on pnpm dev	Root needs packageManager and must not conflict with devEngines.packageManager .
Workspace resolution	name in <code>packages/engine/package.json</code> must match @chess/engine .
Server deps	Include fastify-plugin when using Fastify plugins in this repo.
<code>cd ../../</code> in tutorial	Replaced by "run from monorepo root" so paths are not wrong from another step.